SkillRC: Is It the Skill or the Stack? Auditing Experience Memory in Agent Harnesses

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Abstract

Rules distilled from an agent's past runs can help one model and not another. Is the difference in what the rules say, the skill, or in how each model reads added text, the stack? A control that keeps a memory's items, tags, word counts, order and token count but replaces each rule with a sentence about gardening splits the effect into reading cost and content value. The differences we could attribute lie in the skill. On ALFWorld, no model's reading cost was distinguishable from zero, and all were within 0.04. The content was worth $+0.08$ to GPT-4o-mini on a pool from its own runs, $+0.039$ (95% interval $[0.012,\ 0.065]$) to Qwen3-8B across 18 pools, and nothing measurable to the other four models. On WebShop it lowered the scores of the two readers able to do the task. For Qwen3 readers, rules distilled from experience beat equally long rules written without it by $+0.035$ $[0.014,\ 0.057]$, and the source agent made no difference at balanced coverage. Because only a matched control places an effect in the content, and pools built alike differ about as much as these effects, a memory should be judged against a length-matched control over several pools.

1 Introduction

A common way to let an LLM agent learn from experience is to collect its successful runs, ask a language model to distill them into a short list of rules, and place the list in the agent's prompt (Zhao et al., 2024; Shinn et al., 2023; Wang et al., 2025). The gains are uneven. Evaluations of agent skills find gains that vary across tasks and models and are sometimes negative (Li et al., 2026; Han et al., 2026), and that shrink when skills have to be retrieved under realistic conditions (Liu et al., 2026). In our own runs, the same 33 rules raise GPT-4o-mini's success on ALFWorld by 0.08 and change the success of Llama-3.1-8B, GPT-4.1-mini and Qwen3-14B by less than 0.03. A practitioner cannot tell from such a result whether to change the rules or the model.

A comparison with and without the memory cannot settle this, because adding a memory changes two things at once. The model has more text to read, and the text says something about the task. Language models are sensitive to irrelevant context and to the length and format of their input (Shi et al., 2023; Levy et al., 2024; Sclar et al., 2024), so the first change alone could move success by as much as the gains being reported. We call the effect of the first change the *reading cost* of a memory and the effect of the second its *content value*. Our title asks which of the two differs from one model to the next: the skill, meaning what the memory says, or the stack, meaning the reader model with its tokenizer, chat template and serving. The harness, the layer around the model that builds its context, exposes tools and retrieval, keeps memory and parses outputs (Huang et al., 2026; Lee et al., 2026), is held fixed except for the memory block.

We call our evaluation design SkillRC, a reality check for agent skills. It measures the two parts with a control that keeps everything about the memory except its content. The control has the same number of items with the same tags, each item has the same number of words, the items appear in the same order, and the block has the same token count under every reader's tokenizer, but each rule is replaced by a sentence about vegetable gardening. The control minus no memory is the reading cost, and the memory minus the control is the content value. Two further contrasts ask what the content value consists of. A list of generic rules of the same length, written by a model that saw only a description of the task, tests whether the value needs experience at all. A crossover of the agents that supply the experience with the models that distill it, with three replicate pools for each pairing, tests whether it matters whose experience it is and how much pools built the same way differ.

We find that no reader pays a measurable cost for reading a memory-length block, and that the reader differences we could attribute lie in what the memory says. The content was worth $+0.08$ to GPT-4o-mini, reading a pool built from its own runs, about $+0.04$ to Qwen3-8B across 18 pools, nothing measurable to the other four readers on ALFWorld, and a loss on WebShop for both readers that can do the task. For the Qwen3 readers we tested, distilled rules beat generic rules of the same length, and the agent that supplied the experience made no difference once coverage was equal. Two pools built by the same pipeline can differ by as much as these effects.

Our contribution is an evaluation design for experience memory: a decomposition into reading cost and content value with controls matched in items, order and token count, and audits that check that every task receives the same items in the same order under each condition (§3). We apply it in five studies on ALFWorld (Shridhar et al., 2021) and WebShop (Yao et al., 2022) whose predictions were written down before their runs (§4), with six readers and 24 ALFWorld pools in all.

2 Related Work

Agents that learn from experience store it in many forms: executable skills (Wang et al., 2024), verbal reflections (Shinn et al., 2023), distilled insights and guidelines (Zhao et al., 2024; Fu et al., 2024), reusable workflows (Wang et al., 2025), evolving playbooks (Zhang et al., 2026), and strategies distilled from both successes and failures (Ouyang et al., 2026b). Later systems curate or revise skills around a fixed executor (Yang et al., 2026; Ouyang et al., 2026a; Gao et al., 2026), or optimize the whole harness in which a memory sits (Lee et al., 2026; Huang et al., 2026). Most of this work supports its methods with a gain over the same agent without memory, usually measured on one or two strong models.

Evaluations that look across tasks and models find such gains unstable. SkillsBench reports broad average gains from curated skills together with negative deltas on some tasks, and it notes that an injected skill also adds context, using self-generated skills as a partial control (Li et al., 2026). SWE-Skills-Bench finds limited gains and a token overhead in software engineering (Han et al., 2026), and realistic retrieval erodes skill gains further (Liu et al., 2026). On ALFWorld, abstract procedural memories transfer better than raw trajectories, with negative transfer concentrated on hard cases (Hu et al., 2026), and agents follow retrieved experience closely enough to be misled by it (Xiong et al., 2026). SelfEvoSkill compares executions with and without a skill while holding the agent, task and workspace fixed, and uses the difference to revise the skill (Gao et al., 2026); we use the same kind of matched comparison to evaluate a fixed memory, adding a control for the act of reading. These studies show that the effect of a memory varies, without separating the part due to reading added text.

Prompting research has made that kind of separation for a long time. Content-free inputs calibrate few-shot predictions (Zhao et al., 2021), irrelevant or misleading instructions can work about as well as good ones (Webson and Pavlick, 2022), and demonstrations with random labels keep much of their benefit (Min et al., 2022). Irrelevant context, longer inputs, formatting and position all change what a model does (Shi et al., 2023; Levy et al., 2024; Sclar et al., 2024; Liu et al., 2024). On ALFWorld itself, ReAct performance is driven by the similarity between demonstration tasks and queries more than by the content of the reasoning traces (Verma et al., 2024). Most of these controls were designed for a single prompt. An agent's memory is reread at every step of an episode, by models whose tokenizers and handling of context differ, and its content is distilled from experience that might be replaceable by advice requiring none. We have not found a study that separates reading cost from content value for agent memory across readers, asks whether the content value is specific to experience, or measures how much it changes between pools built by the same pipeline. These are the questions we take up.

3 Method

3.1 Reading cost and content value

An experience memory here is a pool of short rules, each tagged with a task type, distilled from successful trajectories and inserted as one block into the prompt of a frozen model, the *reader*. The harness ranks the items against the current task with BM25 (Robertson and Zaragoza, 2009) and inserts them up to a token budget. In all our studies the whole pool fits, so ranking only sets the order. Nothing else in the prompt changes (Figure 1). For each task t, the realized payload of a condition is the set of pool positions it inserts, I_t, their order σ_t and

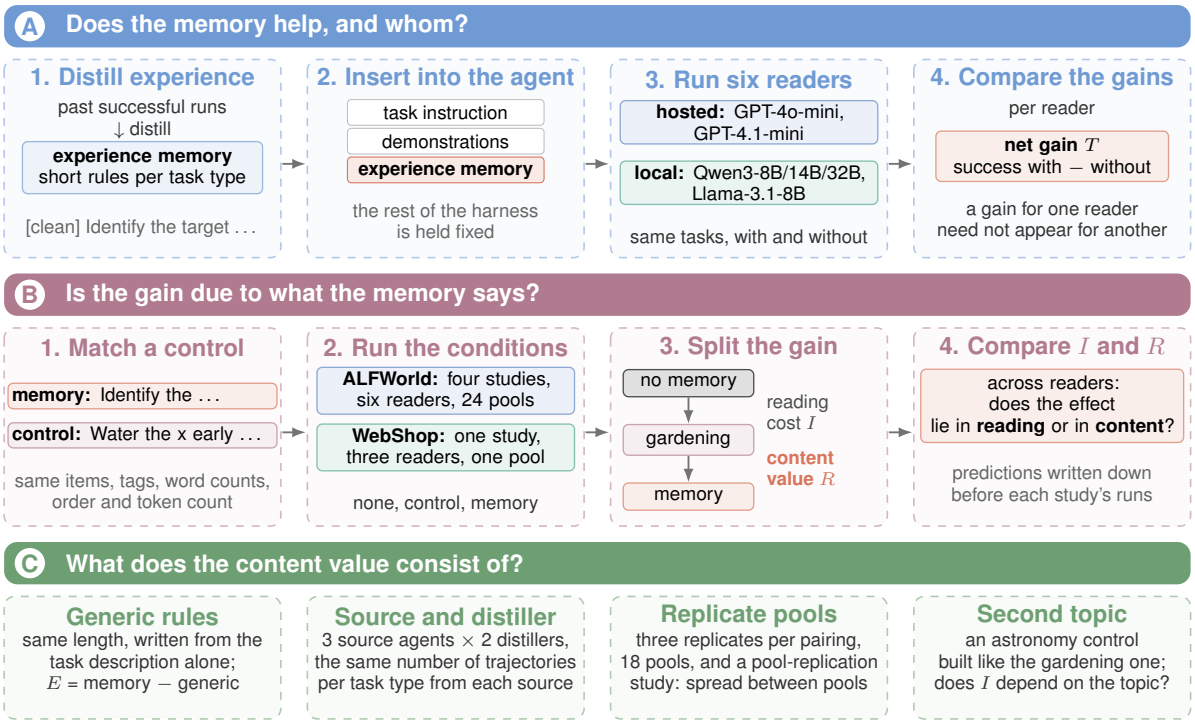


Figure 1: The design. A memory distilled from past runs is inserted as one block into an otherwise fixed agent and read by six models (A). A matched control with the same items, order and token count splits each reader's gain into a reading cost and a content value (B). Further contrasts ask what the content value consists of (C).

their token count ℓ_t. Two conditions are *payload-matched* when I_t and σ_t coincide for every task and ℓ_t is equal under every reader's tokenizer. The harness logs ℓ_t in every episode, and an offline audit replays the retrieval on every task to check I_t and σ_t.

Let $\bar{Y}_{\text{none}}$, $\bar{Y}_{\text{mem}}$ and $\bar{Y}_{\text{irr}}$ be a reader's mean success without memory, with the memory and with the irrelevant control. The net effect splits as

$$\underbrace{\bar{Y}_{\text{mem}} - \bar{Y}_{\text{none}}}_{T,\ \text{net effect}} = \underbrace{\bar{Y}_{\text{irr}} - \bar{Y}_{\text{none}}}_{I,\ \text{reading cost}} + \underbrace{\bar{Y}_{\text{mem}} - \bar{Y}_{\text{irr}}}_{R,\ \text{content value}}. \quad (1)$$

A reading cost can be positive if an added block of any content helps. The split assumes that the cost of reading does not depend on what the text is about, an assumption we return to in Section 4.3. It is also only as good as the control, which is payload-matched to the memory and, as far as we could check, differs from it only in content. Each item keeps its tag and its word count, and its body becomes an imperative sentence about vegetable gardening, drawn from a bank of templates that use no ALFWorld object, receptacle or action word (Table 1). We wanted the control to read like advice. It is ranked with the true pool's BM25 scores and inserted at the same positions. A deterministic pass then replaces the fewest body words with the nonce token x until the block has the memory's token count under the tokenizer of every reader in the study. The 18 crossover controls and the astronomy controls needed no replacement. The focal pool's control, used in the five-reader and generic-rule studies, needed 30 replacements in 472 body words, spread over 19 of its 33 items (for example "Rotate crops every x x then x x then x to rest the tired soil"). The Qwen-sourced pool's control and one pool-replication control needed 3 each, and the WebShop control, about choral singing because WebShop sells garden products, needed 20 in 377 body words.

All estimates in this paper are differences of paired task means. Intervals are 95% percentile intervals from 20,000 bootstrap draws of tasks within task type (product category on WebShop), with pools also resampled where Section 3.2 says so.

3.2 What the content value consists of

Experience or generic advice. If the distilled rules only restated what a model can infer from the task description, a list written without any experience should do as well. For each ALFWorld task type, Qwen3-32B received a fixed description of the environment and of the goal, together with the instruction wording of the distillation prompt,

Condition	Item 3 of the focal pool, tag `[clean]`
Memory	Always clean the item using the designated cleaning area before placing it in the final location.
Generic	To verify the task is complete, carefully examine the object or look at the receptacle closely.
Gardening	Always pull weeds while they are small so they never set seed across the whole garden.
Astronomy	Mount the camera on a sturdy tracking head and begin with short exposures of bright constellations.

Table 1: One item in each condition, copied from the released pools. All four have 16 words, and each whole block has the memory's token count under every reader's tokenizer. This gardening item needed no placeholder.

and wrote six rules without seeing a trajectory. The rules were then paired with the memory's items, rewritten to the same word count and matched in token count and order (Appendix B). The contrast $E = \bar{Y}_{\text{mem}} - \bar{Y}_{\text{gen}}$ is what experience adds beyond generic advice, and $G = \bar{Y}_{\text{gen}} - \bar{Y}_{\text{irr}}$ is what generic advice adds over unrelated text. A second unrelated control about amateur astronomy, built like the gardening one, shows how much the reading cost depends on the topic of the control.

Source, distiller and replicate pools. Pools distilled from different agents differ in coverage as well as in whose experience they hold. On the 140 seen ALFWorld games, GPT-4o-mini succeeded in 73 games, Qwen3-8B in 111 and Qwen3-32B in 116, and GPT-4o-mini solved only one heat game. A pool built from all successes therefore confounds the source with how much of the task its trajectories cover. We drew the same number of successful trajectories per task type from each source, six per type except one for heat, 31 in all. Each source has three replicate subsamples drawn with different seeds, and each subsample was distilled once by Qwen3-32B and once by Qwen3-8B with the same prompt. The result is 18 pools of 31 to 36 rules (660 to 777 tokens), each with its own gardening control. Intervals for pooled estimates resample tasks within type and, for each source, its three subsamples, so they include the variation between pools. A smaller pool-replication study took the same approach with one source, one distiller and one reader: Qwen3-32B distilled four pools from Qwen3-8B's successes, the first six trajectories per type and three random subsamples of six per type, and Qwen3-8B read them.

3.3 Readers, environments and studies

ALFWorld. We use the 134 unseen games with a ReAct scaffold (Yao et al., 2023) that lists the admissible actions, two demonstrations per prompt and a limit of 30 steps. Each run covers every game with three demonstration pairs, 402 episodes. The focal pool holds 33 rules (645 tokens) that GPT-4o-mini distilled from at most six of its own successful trajectories per task type on the seen games. The generic-rule and pool-replication studies also use a second pool, which we call the Qwen-sourced pool: 36 rules (706 tokens) that GPT-4o-mini distilled from Qwen3-8B's successes.

WebShop. WebShop replaces rooms and objects with search and product pages, so it tests whether the decomposition carries over to a different action space. We use the 1,000-product environment with a server that seeds its random generation of prices, so that every session has the same instruction across restarts. The 500 test sessions allow 15 steps and are scored in $[0, 1]$. The pool holds 30 rules (524 tokens) that GPT-4o-mini distilled from its own successes on 576 training sessions.

Readers. Our question concerns differences between readers, so the readers span a range of ability, from 0.26 to 0.90 success without memory on ALFWorld. GPT-4o-mini and GPT-4.1-mini are called through the OpenAI API. Llama-3.1-8B (Grattafiori et al., 2024) and Qwen3-8B, Qwen3-14B and Qwen3-32B (Yang et al., 2025), with thinking off, are served locally with vLLM (Kwon et al., 2023). All decode at temperature zero. Table 2 lists the five studies.

Protocols and noise. For each study we wrote down the predictions, estimands and analysis before its runs. In the five-reader study, one addendum (D1–D2) was written after the hosted readers' runs had been analysed (Appendix A). We call these protocol analyses and mark everything chosen after seeing data as exploratory. Appendix A lists every prediction with its outcome. The first two studies also included a scrambled copy of the memory, with each item's words shuffled until no original adjacent pair remained, as a second control. Its effect changed sign between game samples for the same reader, so we report only the gardening control. Repeating a run under identical settings changes the outcome of 17–20% of GPT-4o-mini

Study	Environment	Readers	Memories	Conditions
Five readers	ALFWorld	GPT-4o-mini, GPT-4.1-mini, Llama-3.1-8B, Qwen3-8B, Qwen3-14B	focal pool	none, memory, gardening
WebShop	WebShop	GPT-4o-mini, GPT-4.1-mini, Qwen3-8B	one pool	none, memory, choral singing
Crossover	ALFWorld	Qwen3-8B, Qwen3-14B, Qwen3-32B	18 balanced pools	none, memory, gardening
Generic rules	ALFWorld	Qwen3-8B, Qwen3-14B	focal and Qwen-sourced pools	the above, generic rules, astronomy
Pool replication	ALFWorld	Qwen3-8B	four replicate pools and the Qwen-sourced pool	none, memory, gardening

Table 2: The five studies. Every condition of a study ran for each of its readers in one time window, with 402 episodes per ALFWorld run and 500 sessions per WebShop run. GPT-4o-mini ran every condition of the first two studies twice.

episodes and 4–13% of local-reader episodes, so we read differences of about 0.02 between single runs as noise.

4 Results

4.1 No reader pays a measurable reading cost

With the focal pool, the gardening control changes success by 0.000 for GPT-4o-mini and by -0.017 to -0.040 for the other four readers, and every interval includes zero (Figure 2, left). We had expected the reading cost to explain why a memory helps some readers and not others, because a memory adds several hundred tokens of instructions and models differ in how they handle added context. In an addendum written before the local readers' estimates were computed, the five-reader study therefore predicted that the reading cost would vary across readers more than the content value. That prediction failed. Across the five readers the reading cost spans 0.040 and the content value 0.077. The difference between the two spans is itself uncertain, $[-0.050,\ 0.095]$ (exploratory), and so are the differences between readers' reading costs, whose intervals reach 0.10. For GPT-4.1-mini minus GPT-4o-mini, for instance, the difference is -0.040 $[-0.100,\ 0.020]$.

The ± 0.04 band was set from the range of these five estimates and then used as a prediction for the crossover, so for the five readers above it is a description, not a test. The intervals rule out reading costs beyond ± 0.04 only for GPT-4o-mini averaged over its two runs, $[-0.037,\ 0.031]$, and for Qwen3-32B. In the crossover, averaged over pools, the reading cost is $+0.006$ for Qwen3-8B, $+0.020$ for Qwen3-14B and -0.010 for Qwen3-32B, inside the predicted band, and no single pool's control moves any of the three readers by more than 0.037. Across all studies the largest single-pool estimate is -0.042. On WebShop the choral control changes the score by $+0.018$ for GPT-4o-mini and -0.016 for Qwen3-8B (Table 3).

The topic of the unrelated text matters more than we expected. The generic-rule study predicted that replacing gardening with amateur astronomy would shift the reading cost by less than 0.03, with an interval including zero. The shift was 0.025 $[0.007,\ 0.044]$, small but distinguishable from zero, so the reading cost is $+0.017$ $[-0.010,\ 0.045]$ under one topic and -0.007 $[-0.034,\ 0.019]$ under the other. Measured against the astronomy control, Qwen3-8B's content value in that study would be about $+0.075$ instead of $+0.044$ (exploratory), and absolute content values carry a margin of this size.

4.2 Two readers gain reliably from the content

Content value is reliably positive for two readers, GPT-4o-mini with the focal pool and Qwen3-8B across 18 pools. With the focal pool, content values range from $+0.002$ to $+0.080$, and only GPT-4o-mini's excludes zero, at $+0.080$ $[0.030,\ 0.129]$ in the protocol run and $+0.091$ $[0.050,\ 0.133]$ averaged over its two runs. On the 74 games added after the first 60, it is $+0.059$ $[-0.009,\ 0.126]$. Qwen3-8B follows at $+0.040$, GPT-4.1-mini at $+0.032$, Qwen3-14B at $+0.012$ and Llama-3.1-8B at $+0.002$. GPT-4o-mini's content value exceeds Llama-3.1-8B's by 0.077 $[0.012,\ 0.142]$ and Qwen3-14B's by 0.067 $[0.012,\ 0.122]$, unadjusted for multiplicity. Against GPT-4.1-mini, the gap in net effect, 0.087 $[0.025,\ 0.152]$, splits about evenly between reading cost (0.040) and content value (0.047), neither of them reliable on its own. GPT-

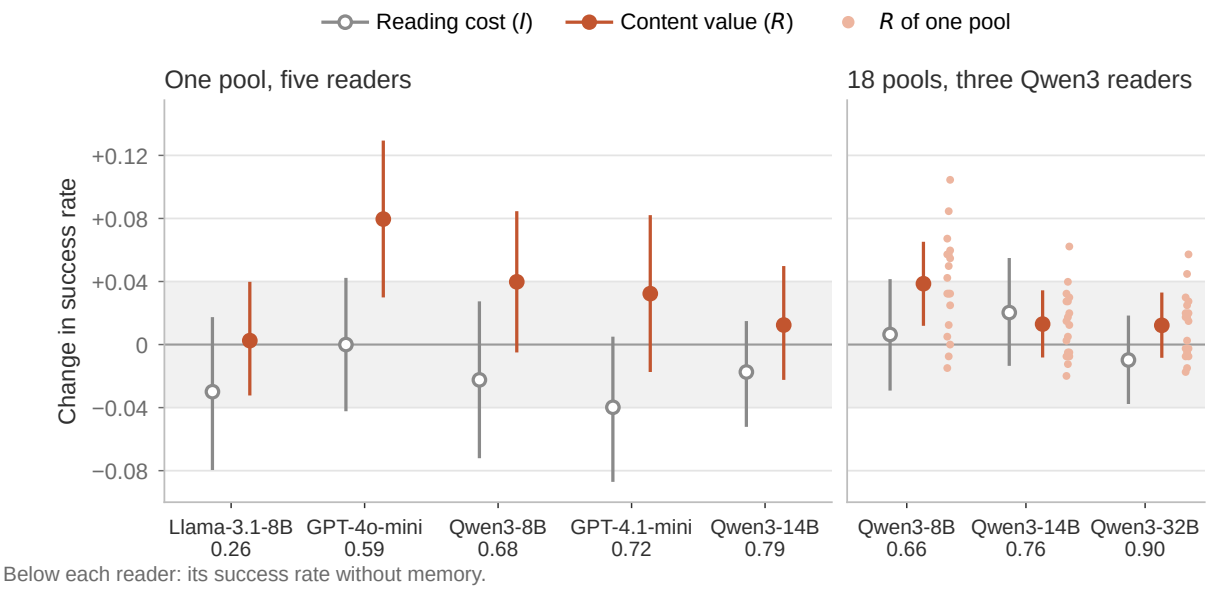


Figure 2: Reading cost I (unrelated text minus no memory) and content value R (memory minus unrelated text) on 134 unseen ALFWorld games, with 95% intervals. Left: the focal pool read by five readers (single runs). Right: the mean over 18 pools for three Qwen3 readers, with intervals that resample pools as well as tasks. Light dots are the content values of the individual pools. The shaded band marks ± 0.04, the bound predicted for the crossover.

4o-mini is also the agent whose trajectories, distilled by GPT-4o-mini, make up the focal pool, so its lead may partly be an own-source advantage, which the crossover could not resolve (§4.3).

Over 18 pools (Figure 2, right), the content value is $+0.039$ $[0.012, 0.065]$ for Qwen3-8B, $+0.013$ $[-0.008, 0.034]$ for Qwen3-14B and $+0.012$ $[-0.008, 0.033]$ for Qwen3-32B. The crossover's protocol predicted that Qwen3-8B would gain reliably more than Qwen3-32B. The difference, $+0.026$ $[-0.007, 0.060]$, does not meet that bar (Holm-adjusted $p = 0.52$), so the ordering by size remains a trend.

Exploratory: content value was reliable only for two mid-range readers, whose success without memory is 0.59 and 0.66–0.68; we note this pattern without testing it. Llama-3.1-8B solves a quarter of the games without memory and gains nothing. If each GPT-4o-mini episode is classified by how it did on the same game in the other two no-memory episodes, its content value is $+0.196$ where it failed both and $+0.006$ where it solved both, a difference of $+0.190$ $[0.076, 0.309]$. Llama-3.1-8B gains nothing even on the games it fails.

4.3 Distilled rules beat generic rules, and pools differ as much as the effects

In the generic-rule study, read by Qwen3-8B and Qwen3-14B, experience rules beat generic rules of the same length by $+0.035$ $[0.014, 0.057]$ over two pools and two readers, as predicted, with $+0.058$ for Qwen3-8B and $+0.012$ for Qwen3-14B (Figure 3). The study also predicted that the generic rules would beat the gardening text, which they did not, at -0.016 $[-0.037, 0.005]$. About half of the gap is therefore the content value of the experience rules in this study, $+0.019$ $[-0.002, 0.040]$, and the rest is the generic rules doing slightly worse than unrelated text. The gap is not explained by greater word overlap with the test games: the generic rules share more words with them than the distilled pools do, 8.5–9.3% of their words against 3.9–5.5%. The two lists also differ in their writer. The experience pools were distilled by GPT-4o-mini and the generic rules written by Qwen3-32B. On the focal pool, Qwen3-8B's advantage for experience comes entirely from generic rules doing worse than gardening, -0.062 $[-0.104, -0.020]$. Text about the task may cost more to read than text about gardening, in which case part of what we call content value is a reading cost that depends on content.

The agent that supplied the experience made no difference once coverage was equal. With six trajectories per type from each source (one for heat), the content value of pools built from Qwen3-8B's successes and from GPT-4o-mini's differs by -0.002 $[-0.024, 0.019]$ for the three Qwen3 readers, as predicted, and Qwen3-

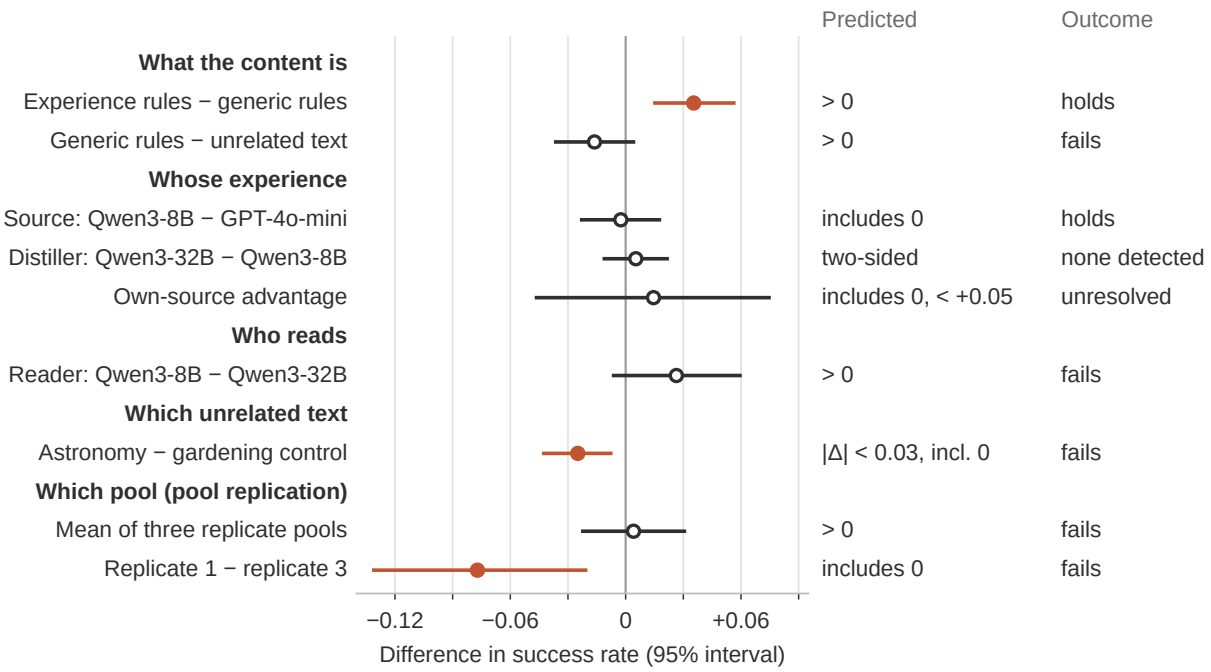


Figure 3: Protocol contrasts of the generic-rule study (pooled over two pools and two readers), the crossover (pooled over pools and three readers) and the pool-replication study (reader Qwen3-8B), each with its prediction and outcome. Filled markers have intervals that exclude zero.

32B's pools differ from Qwen3-8B's by -0.006 $[-0.029, 0.015]$. Qwen3-32B and Qwen3-8B distillations of the same subsamples differ by $+0.005$ $[-0.012, 0.022]$, so the distilling model made no detectable difference either. Whether a reader gains more from its own experience is unresolved, at $+0.015$ $[-0.047, 0.075]$. These results concern what a trajectory contributes for Qwen3 readers. A more successful agent still leaves more trajectories, and the balanced design removes that advantage by construction.

For a fixed source, distiller and reader, the crossover's three replicate pools differ by more than task sampling explains ($Q = 72.2$ on 36 degrees of freedom, $p = 0.0003$). The between-pool standard deviation of 0.019 is close to the 0.02 we had predicted and comparable to the content values themselves. For Qwen3-8B, single pools range from -0.015 to $+0.105$. Two distillations of the same subsample agree only moderately, with a correlation of 0.42 over 27 pairs, so both the choice of trajectories and the distillation contribute. Single-pool content values should be read with this spread in mind, and the pooled intervals in this section and in Figure 2 resample pools as well as tasks.

The pool-replication study points the same way (Table 8). Its three random subsamples gave Qwen3-8B content values of -0.025, -0.015 and $+0.052$, a mean of $+0.004$ $[-0.023, 0.032]$, and the pool from the first six trajectories per type gave $+0.032$. The crossover pools with the same source, distiller and reader gave $+0.041$ $[0.005, 0.077]$, though they kept one heat trajectory where these kept six. The study had predicted that its pools would carry value and agree with one another. Both predictions failed: the first and third subsamples differ by -0.077 $[-0.132, -0.020]$.

4.4 On WebShop the distilled rules lower both capable readers' scores

On WebShop the content value is negative (Table 3). The pool lowers GPT-4o-mini's score from 0.294 to 0.154 while the choral control leaves it at 0.312, a content value of -0.158 $[-0.191, -0.125]$, and

Reader	none	choral	memory	I	R
GPT-4o-mini (1)	0.294	0.312	0.154	$+0.018$	-0.158
GPT-4o-mini (2)	0.303	0.323	0.167	$+0.020$	-0.156
Qwen3-8B	0.329	0.313	0.198	-0.016	-0.116
GPT-4.1-mini	0.029	0.028	0.027	-0.001	-0.002

Table 3: WebShop, 500 test sessions: mean score per condition, reading cost I and content value R, with GPT-4o-mini's two runs. Every R interval for GPT-4o-mini and Qwen3-8B excludes zero and every I interval includes it (Appendix C). GPT-4.1-mini fails almost every session in every condition.

a second run gives -0.156. For Qwen3-8B it is -0.116 $[-0.153, -0.079]$. The protocol had predicted a positive value for GPT-4o-mini. GPT-4.1-mini fails nearly every session in every condition, so its estimates carry no information; in an exploratory reading of its sessions, it typically opens one of a product's sub-pages, which have no Buy button, and keeps clicking there until the step limit. In an exploratory count of actions, the pool raises GPT-4o-mini's searches per session from 2.44 to 5.69, against 2.35 with the control, and Qwen3-8B's from 1.88 to 2.71. Several rules tell the agent to search again, such as "If the desired product is not found, consider re-searching with the same criteria" for fashion, the category of most test sessions. GPT-4o-mini, the reader with the largest content value on ALFWorld, has the largest loss here, and its reading cost is near zero in both environments.

5 Conclusion

The differences between models that we could attribute lie in the skill, in what the memory says. Reading a memory of 520 to 790 tokens cost no model we tested a measurable amount, while what the rules said was worth $+0.08$ to GPT-4o-mini on a pool from its own runs, about $+0.04$ to Qwen3-8B, nothing measurable to the other four readers on ALFWorld, and a loss on WebShop.

Where a memory did nothing for a model, as for Llama-3.1-8B and Qwen3-14B with the focal pool and Qwen3-32B across 18 pools, its content value was near zero and so was its reading cost, so no cost of reading was masking a gain. GPT-4o-mini's lead over Llama-3.1-8B and Qwen3-14B lies in content value, with no detectable difference in reading cost. On WebShop the distilled rules lowered the scores of the two readers that can do the task. For Qwen3-8B and Qwen3-14B, the readers we also tested with generic advice, distilled rules beat generic rules of the same length, and the generic rules were worth no more than unrelated text; for all three Qwen3 readers, the agent that supplied the experience made no difference once coverage was equal. A memory should therefore be judged against a length-matched control, which is what places an effect in the content, and over several pools, since two pools built by the same pipeline can differ by as much as the effect itself.

Limitations

Our readers are open models of 8B to 32B parameters and two small hosted models, and our memories are rule lists of 520 to 790 tokens. Long procedural skills and frontier models may behave differently. Each reader is a whole stack of weights, tokenizer, chat template, serving and decoding, so differences between readers are differences between stacks and say nothing about the weights alone. Both environments are small, with 134 unseen ALFWorld games and a 1,000-product WebShop with synthetic instructions.

The decomposition depends on its controls and on the assumption that reading cost does not depend on content. Text about the task may cost more to read than text about gardening, which would move part of the reading cost into what we call content value; the generic rules doing worse than gardening is consistent with this. The unrelated text may also differ from a memory in style and plausibility, two of its controls carry placeholder tokens, and the topic of the unrelated text moves the reading cost (§4.1). The generic rules were written by a different model from the experience rules, and forcing them to exact lengths added filler to some of them, so a stronger writer might produce more useful generic advice.

In the crossover, heat tasks rest on a single trajectory per pool, and the replicate subsamples overlap (GPT-4o-mini's single heat trajectory is in all three of its subsamples), so the between-pool standard deviation is a lower bound for independent pools. The hosted readers did not take part, and the own-source and reader contrasts are underpowered. The five-reader and WebShop studies use one pool each, so given the variation between pools we measured, their content values are single draws, and GPT-4o-mini's content value there may include an own-source advantage. We measured how much each reader gains from the content and did not establish why readers differ. The hosted models were reached through aliases that their provider can change. The exploratory analyses listed in Appendix A were chosen after the data were seen, and only the four main crossover contrasts were adjusted for multiplicity.

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A Protocols and outcomes

Table 4 lists every prediction written down before the runs of the five studies, with its outcome. The five-reader study's comparison of spans (D1) and its prediction test (D2) were added in an addendum after the hosted readers' runs had been analysed and before any estimate from the local readers was computed, so for the hosted readers they are descriptive. One deviation occurred, before any episode of the generic-rule study: single-shot rewrites to an exact word count came out three to four words short, so they were replaced by a loop that told the writer how many words to add or remove. The writer still saw only the task description and its own base rule. In the crossover study, the four contrasts named in Table 6 were Holm-adjusted as planned, and all adjusted p-values are at least 0.52.

B Controls and generic rules

Unrelated text. Each control is built from a bank of template sentences on one topic, with every word checked against the text of the test games and the vocabulary of the environment and pools. For each item, the builder picks a sentence with the item's word count. A greedy pass then replaces the fewest body words with the nonce x until the block's token count equals the memory's under the tokenizer of every reader in the study: o200k for the hosted models, and the Llama-3.1 and Qwen3 tokenizers, the latter byte-identical across Qwen3-8B, Qwen3-14B and Qwen3-32B. Section 3.1 gives the number of replacements per control. For the crossover, generic-rule and pool-replication studies, an audit replayed retrieval on the first observation of all 134 games for every memory and control pair and found no difference in order, item set or token count under any tokenizer.

Study	Prediction	Outcome
Five readers	P1, A3: net gain for GPT-4o-mini, none detectable for Qwen3-8B	hold
	Predictions about the scrambled copy (P2–P4 and A1, A2, A4 on the two game batches; C1, C2 exploratory)	P2, P3, C1, C2 fail; A1 fails for GPT-4o-mini ($+0.009$) and holds for Qwen3-8B; P4, A2, A4 hold
	B1: under 25% of GPT-4o-mini episodes change between runs, signs stable	holds (17–20%)
	D1: the reading cost spans more across readers than the content value	fails (0.040 against 0.077)
	D2: reading cost plus the others' mean content value predicts net gain better than the others' mean net gain	holds, weakly (error 0.026 against 0.037)
WebShop	W1: positive content value for GPT-4o-mini; W2: reading cost spans more than content value; W3: the unrelated text does no better than the scrambled copy; W4: GPT-4.1-mini gains less than GPT-4o-mini	all fail; content value is negative for GPT-4o-mini and Qwen3-8B
Crossover	A1: no own-source advantage, interval below $+0.05$	unresolved ($+0.015$ $[-0.047,\ 0.075]$)
	A2: Qwen3-8B gains more than Qwen3-32B	fails ($+0.026$ $[-0.007,\ 0.060]$)
	A3: at balanced counts the Qwen3-8B and GPT-4o-mini sources do not differ	holds (-0.002)
	A4: distiller effect, two-sided	none detected ($+0.005$)
	A5: pools differ ($p < 0.05$) with between-pool SD at least 0.02	partly ($p = 0.0003$, SD 0.019)
	A6: reading cost within ± 0.04 for each reader	holds for all three
Generic rules	B1: experience rules beat generic rules	holds ($+0.035$)
	B2: generic rules beat unrelated text	fails (-0.016)
	B3: the topic of the unrelated text changes the reading cost by less than 0.03, interval including zero	fails (-0.025 $[-0.044,\ -0.007]$)
Pool replication	P1: replicate pools carry content value; P2: their range is below 0.05; P3: no pairwise difference; P4: reading cost within ± 0.04	P1–P3 fail (mean $+0.004$, range 0.077, R1 $-$ R3 $= -0.077$ $[-0.132,\ -0.020]$); P4 holds (-0.015)

Table 4: Predictions written down before the runs of the five studies and their outcomes. Exploratory analyses in the paper are the WebShop action counts, the reading of GPT-4.1-mini's WebShop sessions, the comparison of readers by success without memory, the split by difficulty and its interval, the interval for the difference of spans, and the content value against the astronomy control. The five-reader gaps between readers are protocol estimands without predictions and are not adjusted for multiplicity.

Generic rules. Qwen3-32B (thinking off, temperature 0) wrote six rules per task type from the prompt below. For the clean type, `{goal}` reads: the goal reads like "put a clean [object] in [receptacle]" or "clean some [object] and put it in [receptacle]".

> ALFWorld is a text-based household environment. At the start of an episode the agent is told which receptacles it can see in the room (for example cabinets, drawers, countertops, shelves, a fridge, a microwave, a sinkbasin, a stoveburner, a garbagecan, a desklamp), each with a number. Objects lie on or inside receptacles, and closed receptacles must be opened to see inside. The agent acts with text commands: go to [receptacle], open [receptacle], close [receptacle], take [object] from [receptacle], put [object] in/on [receptacle], clean [object] with [receptacle], heat [object] with [receptacle], cool [object] with [receptacle], use [object], examine [object or receptacle], look, inventory. An episode has a limited number of steps.
>
> Task type '{task_type}': {goal}
>
> Write {n} concise, GENERAL rules (heuristics) that would help an agent solve this task type. Each rule: one imperative line, no specific object numbers or room layouts. Output ONLY a bullet list, each line starting with '- '.

The last two sentences copy the distillation prompt (Appendix D). Within each type the rules whose word counts were closest to the memory's items were paired with them. Each rule was then rewritten to the item's exact word count, keeping its meaning, and the variants were chosen to match the memory's token count under both tokenizers, with no nonce replacement needed. In 13–20% of episodes the first action differs between the generic and the experience rules.

C Further results

Tables 5, 6, 7 and 8 give the estimates behind Figures 2 and 3. On WebShop the intervals for GPT-4o-mini (run 1) are $[-0.012,\ 0.048]$ for I and $[-0.191,\ -0.125]$ for R, and for Qwen3-8B

$[-0.056, 0.024]$ and $[-0.153, -0.079]$. Between the two GPT-4o-mini runs, 13–17% of sessions change score.

D Setup details

Models. The GPT-4o-mini alias resolved to the snapshot of 2024-07-18, and GPT-4.1-mini was called as the snapshot of 2025-04-14. The local models were served with vLLM in bf16 with a context of 16,384 tokens, at these Hugging Face revisions: Qwen3-8B `b968826d`, Qwen3-14B `40c06982`, Qwen3-32B `9216db57` (tensor parallelism over two GPUs), and Llama-3.1-8B-Instruct from the `unsloth` mirror, `4699cc75`. The Qwen3 models ran with `enable_thinking=false`. All calls used temperature 0 and at most 256 completion tokens.

ALFWorld. Appendix E gives the prompts and loop rules, and Appendix F the task types and the focal memory. Demonstration pairs are selected by seeds 0, 1 and 5. The released code lists the 134 unseen games in task-id order. The sources of the crossover ran without memory on the 140 seen games with one demonstration pair; their heat successes numbered 1 (GPT-4o-mini), 14 (Qwen3-8B) and 16 (Qwen3-32B). Pools were distilled with one call per task type, at temperature 0 and up to 512 tokens, from the prompt below, where `{trajs}` holds up to six serialized successful trajectories of the type.

> Below are successful agent trajectories for ALFWorld '{task_type}' tasks. Extract 3 to 6 concise, GENERAL insights (rules/heuristics) that help solve this task type. Each insight: one imperative line, no specific object numbers or room layouts. Output ONLY a bullet list, each line starting with '- '. Trajectories: {trajs}

WebShop. We use the official code and 1,000-product data (Yao et al., 2022). Test sessions 0–499 fall into five product categories (fashion 393, garden 57, beauty 23, electronics 19, grocery 8), and the bootstrap stratifies by category. The agent writes a Thought line and an Action line at each step and sees the valid actions (Appendix E). The pool was distilled with the ALFWorld prompt, with the environment words changed, from the first six successful trajectories per category (five for garden and grocery).

Artifacts. ALFWorld and WebShop are released under the MIT license, Qwen3 under Apache 2.0 and Llama 3.1 under its community license. Our harness, configurations, pools, controls, per-episode outcomes, result files and analysis code are released under the Apache License 2.0 at the repository named on the first page.

E Agent prompts

These are the prompts of every condition, as built by `skillrc/react.py`. A memory or control condition adds only the block headed `# Retrieved experience / skills`, which holds the inserted items with their type tags, in BM25 order and separated by blank lines. Without memory the block is absent. Angle brackets mark text filled in at run time.

ALFWorld, system message

```
Interact with a household to solve a task. Output only the
      next line: either an action, or a line beginning with
      'think:' to reason (a 'think:' line is not an
      environment action). Here are two examples.

<first ReAct demonstration for the task type>

<second ReAct demonstration for the task type>

Here is the task.

# Retrieved experience / skills
<memory items>
```

ALFWorld, user message

```
<first observation and task, then for each earlier reply a
      line "> reply" and its observation>
Valid actions right now: <admissible actions, separated by
      "; ">

>
```

The demonstrations are the ReAct prompts of Yao et al. (2023) for the task type, two of the three chosen by the seed. The model's reply is parsed as its first non-empty line, with a leading `>` removed (`look` if the reply is empty). A line that begins with `think:` is added to the transcript with the answer `OK.` and costs no environment step. Any other line is sent to the environment, and a line `put X in Y`, `put X on Y` or `put X in/on Y` is first rewritten to `move X to Y`, the form the installed ALFWorld accepts. An episode ends at success, after 30 environment steps or after 60 model calls, whichever comes first.

WebShop, system message

```
You are an agent solving an interactive text task using the
      ReAct framework.
At each step: first write one short line starting with
      'Thought:' with your reasoning, then output exactly
      one line starting with 'Action:' containing a single
      admissible action. Output nothing after the Action
      line.

# Examples
Task: Find a red cotton shirt under $30.
Thought: I should search using the requested attributes.
Action: search[red cotton shirt]
```

Reader	none	Reading cost I	Content value R	Net effect T
GPT-4o-mini, run 1	0.595	$0.000\ [-0.042,\ 0.042]$	$+0.080\ [0.030,\ 0.129]$	$+0.080\ [0.027,\ 0.132]$
GPT-4o-mini, run 2	0.592	$-0.005\ [-0.050,\ 0.040]$	$+0.102\ [0.050,\ 0.154]$	$+0.097\ [0.047,\ 0.147]$
GPT-4o-mini, mean of runs	0.593	$-0.002\ [-0.037,\ 0.031]$	$+0.091\ [0.050,\ 0.133]$	$+0.088\ [0.050,\ 0.128]$
Qwen3-8B	0.677	$-0.022\ [-0.072,\ 0.027]$	$+0.040\ [-0.005,\ 0.085]$	$+0.017\ [-0.030,\ 0.065]$
GPT-4.1-mini	0.721	$-0.040\ [-0.087,\ 0.005]$	$+0.032\ [-0.017,\ 0.082]$	$-0.007\ [-0.052,\ 0.037]$
Llama-3.1-8B	0.259	$-0.030\ [-0.080,\ 0.017]$	$+0.002\ [-0.032,\ 0.040]$	$-0.027\ [-0.072,\ 0.015]$
Qwen3-14B	0.794	$-0.017\ [-0.052,\ 0.015]$	$+0.012\ [-0.022,\ 0.050]$	$-0.005\ [-0.042,\ 0.032]$

Table 5: Five-reader study, focal pool, 134 unseen games. GPT-4o-mini and Qwen3-8B ran in two same-day batches.

Source	Distiller	Qwen3-8B reads	Qwen3-14B reads	Qwen3-32B reads
GPT-4o-mini	Qwen3-32B	$+0.055\ [-0.005,\ 0.114]$	$+0.022\ [-0.006,\ 0.050]$	$+0.010\ [-0.028,\ 0.047]$
GPT-4o-mini	Qwen3-8B	$+0.056\ [0.009,\ 0.101]$	$+0.005\ [-0.033,\ 0.045]$	$+0.002\ [-0.028,\ 0.032]$
Qwen3-8B	Qwen3-32B	$+0.041\ [0.005,\ 0.077]$	$-0.003\ [-0.036,\ 0.030]$	$+0.019\ [-0.022,\ 0.059]$
Qwen3-8B	Qwen3-8B	$+0.035\ [-0.012,\ 0.084]$	$+0.032\ [0.000,\ 0.064]$	$+0.012\ [-0.034,\ 0.061]$
Qwen3-32B	Qwen3-32B	$+0.027\ [-0.022,\ 0.075]$	$+0.020\ [-0.026,\ 0.070]$	$+0.025\ [0.001,\ 0.051]$
Qwen3-32B	Qwen3-8B	$+0.018\ [-0.030,\ 0.067]$	$+0.003\ [-0.032,\ 0.039]$	$+0.005\ [-0.025,\ 0.036]$
All 18 pools: content value R		$+0.039\ [0.012,\ 0.065]$	$+0.013\ [-0.008,\ 0.034]$	$+0.012\ [-0.008,\ 0.033]$
All 18 pools: reading cost I		$+0.006\ [-0.029,\ 0.041]$	$+0.020\ [-0.014,\ 0.055]$	$-0.010\ [-0.038,\ 0.018]$
All 18 pools: net effect T		$+0.045\ [0.009,\ 0.080]$	$+0.033\ [0.000,\ 0.068]$	$+0.002\ [-0.021,\ 0.027]$
Success without memory		0.657	0.764	0.898

Table 6: Crossover study: content value per source and distiller (mean of three replicate pools) for each reader, with intervals that resample pools and tasks. Holm-adjusted p-values for the four planned contrasts (reader Qwen3-8B minus Qwen3-32B, source Qwen3-8B minus GPT-4o-mini, distiller, own source) are 0.52, 1, 1 and 1. Two identical runs of one pool differ on 12%, 4% and 5% of episodes for the three readers.

Reader	Pool	gardening	astronomy	generic	memory	E	G
Qwen3-8B	focal	0.687	0.652	0.624	0.697	$+0.072\ [0.027,\ 0.117]$	$-0.062\ [-0.104,\ -0.020]$
Qwen3-8B	Qwen-sourced	0.642	0.614	0.674	0.719	$+0.045\ [0.002,\ 0.087]$	$+0.032\ [-0.017,\ 0.082]$
Qwen3-14B	focal	0.786	0.784	0.776	0.776	$0.000\ [-0.030,\ 0.030]$	$-0.010\ [-0.047,\ 0.027]$
Qwen3-14B	Qwen-sourced	0.796	0.761	0.771	0.796	$+0.025\ [-0.007,\ 0.057]$	$-0.025\ [-0.055,\ 0.005]$
Pooled over the four cells						$+0.035\ [0.014,\ 0.057]$	$-0.016\ [-0.037,\ 0.005]$

Table 7: Generic-rule study: success per condition and the two protocol contrasts, E (memory minus generic rules) and G (generic rules minus gardening). Success without memory is 0.657 for Qwen3-8B and 0.764 for Qwen3-14B. Pooled over the four cells, the content value against gardening is $+0.019\ [-0.002,\ 0.040]$, the reading cost is $+0.017\ [-0.010,\ 0.045]$ under gardening and $-0.007\ [-0.034,\ 0.019]$ under astronomy, and astronomy minus gardening is $-0.025\ [-0.044,\ -0.007]$.

```
Observation: Search results include product ids.
Thought: I should inspect the most relevant product.
Action: click[b012example]
Observation: Product page with options and Buy Now.
Thought: I should select the requested option before buying.
Action: click[red]
Observation: Red is selected.
Thought: The item matches, so I should purchase it.
Action: click[buy now]

Task: Find a 32 ounce insulated water bottle below $25.
Thought: I should search for the product and key constraint.
Action: search[32 ounce insulated water bottle]
Observation: Search results include product ids.
Thought: I should inspect a plausible result.
Action: click[b045example]
Observation: Product page with size options.
Thought: I need the requested size.
Action: click[32 oz]
Observation: 32 oz is selected.
Thought: The constraints match, so I should purchase it.
Action: click[buy now]
```

```
# Retrieved experience / skills
<memory items>
```

WebShop, user message

```
<first observation, which includes the instruction>

# Valid actions right now
<admissible actions, separated by "; ">

# Interaction so far
<earlier steps as "Action: ..." and "Observation: ..."
      lines, or "(none yet)">

What is your next Thought and Action?
```

The reply is parsed by its last `Action:` line, or its last non-empty line if it has none. A session ends at a purchase or after 15 steps.

Pool	Rules	Tokens	Content value R	Reading cost I
Qwen-sourced pool (distilled by GPT-4o-mini)	36	706	$+0.067$ $[0.020, 0.114]$	-0.040 $[-0.085, 0.005]$
First six trajectories per type	33	756	$+0.032$ $[-0.017, 0.085]$	-0.022 $[-0.067, 0.022]$
Random subsample 1	36	773	-0.025 $[-0.067, 0.017]$	$+0.015$ $[-0.027, 0.055]$
Random subsample 2	36	790	-0.015 $[-0.060, 0.027]$	-0.017 $[-0.067, 0.032]$
Random subsample 3	36	752	$+0.052$ $[0.005, 0.100]$	-0.042 $[-0.092, 0.005]$
Mean of the three subsamples			$+0.004$ $[-0.023, 0.032]$	-0.015 $[-0.055, 0.024]$

Table 8: Pool-replication study, reader Qwen3-8B, 134 unseen games; success without memory 0.692. All pools except the first were distilled by Qwen3-32B from Qwen3-8B's successes on the seen games, with six trajectories per type. Subsamples 1 and 3 differ by -0.077 $[-0.132, -0.020]$. Two identical runs of subsample 1 differ on 13% of episodes.

F Memories and task types

Table 9 lists the six ALFWorld task types. Tables 10 and 11 reproduce the focal and WebShop memories verbatim from the released pool files, in pool order.

Type	Goal	Games	Rules
`clean`	clean an object, then put it in a receptacle	31	5
`cool`	cool an object, then put it in a receptacle	21	6
`examine`	look at an object under the desk lamp	18	6
`heat`	heat an object, then put it in a receptacle	23	5
`put`	put an object in a receptacle	24	6
`puttwo`	put two objects of one kind in a receptacle	17	5
All		134	33

Table 9: ALFWorld task types, with the number of the 134 unseen games of each type and the number of rules of that type in the focal memory.

#	Type	Rule
1	clean	Identify the target item and its cleaning requirements before starting the task.
2	clean	Prioritize checking locations where the target item is likely to be found, such as countertops, cabinets, and drawers.
3	clean	Always clean the item using the designated cleaning area before placing it in the final location.
4	clean	Confirm the cleanliness of the item if necessary before moving it to the final destination.
5	clean	Ensure to follow the sequence of actions: find, clean, and then place the item correctly.
6	cool	Start by identifying likely locations for the target item based on its type.
7	cool	Always check the fridge first for items that need to be cooled.
8	cool	If the desired item is not found, systematically explore other potential locations.
9	cool	After obtaining the item, ensure to cool it before proceeding to the final placement.
10	cool	Confirm the availability of the final destination before moving the item there.
11	cool	Maintain a clear sequence of actions: find, cool, and place the item.
12	examine	Identify the required items and prioritize searching in likely locations based on their common placements.
13	examine	Check multiple potential locations systematically if the first search does not yield results.
14	examine	Use the most probable locations for the second required item after obtaining the first.
15	examine	Always confirm the presence of an item before attempting to use it.
16	examine	If an item is not found in one location, move to the next logical location without skipping options.
17	examine	Ensure to use the item in the correct context to achieve the task's goal.
18	heat	Identify the target item and its required state before proceeding with the task.
19	heat	Prioritize checking locations where the target item is likely to be found.
20	heat	Use appropriate appliances or tools to modify the item as needed before completing the task.
21	heat	Ensure to open or interact with containers before placing or retrieving items.
22	heat	Maintain a clear sequence of actions to avoid unnecessary backtracking.
23	put	Start by checking locations where the target object is likely to be found, such as tables, countertops, or drawers.
24	put	If the initial location does not contain the target object, systematically explore other nearby locations.
25	put	Always check closed containers (drawers, cabinets) before assuming the object is not present.
26	put	Once the target object is found, plan the route to the designated location for placing it.
27	put	Ensure to confirm the target object is in your possession before attempting to place it.
28	put	When placing the object, verify that the destination is clear and appropriate for the item.
29	puttwo	Start by identifying the most likely locations for the target items based on their type.
30	puttwo	Check each location systematically, ensuring to open closed containers before searching inside.
31	puttwo	Once an item is found, immediately plan the next steps to retrieve and place it in the designated location.
32	puttwo	If the first search does not yield results, revisit previously checked locations, as items may be present in multiple instances.
33	puttwo	Keep track of items already found and placed to avoid unnecessary repetition in searches.

Table 10: The focal memory (`data/pools/expel_insights.jsonl`): 33 rules, 645 tokens, which GPT-4o-mini distilled from its own successful trajectories on the seen games. A memory condition with this pool inserts all 33 rules, and BM25 sets only their order.

#	Category	Rule
1	beauty	Always include specific attributes and price limits in your search queries.
2	beauty	Use concise and relevant keywords to improve search accuracy.
3	beauty	Click on products that match your criteria to verify details before purchasing.
4	beauty	Utilize pagination to explore multiple pages of search results if necessary.
5	beauty	Confirm product specifications and features before finalizing the purchase.
6	beauty	Ensure the product meets all specified requirements to avoid returns.
7	electronics	Use specific keywords that describe the product features and requirements.
8	electronics	Include price constraints in the search query to filter results effectively.
9	electronics	Click on the first relevant product to streamline the purchasing process.
10	electronics	Ensure to check product details for compliance with specified features before buying.
11	electronics	Utilize pagination to explore more options if initial results are unsatisfactory.
12	electronics	Confirm product attributes such as color and style during the selection process.
13	fashion	Use specific keywords related to the product features when searching.
14	fashion	Click on the relevant filters for size and color after the initial search.
15	fashion	Always check the price range to ensure it meets the budget criteria.
16	fashion	If the desired product is not found, consider re-searching with the same criteria.
17	fashion	Prioritize clicking on the product that matches all specified attributes.
18	fashion	Complete the purchase by confirming the selected options before finalizing.
19	garden	Always include specific attributes like color, size, and price in your search queries to narrow down results effectively.
20	garden	Utilize filters for size and color after initial search results to refine options further.
21	garden	If the first search does not yield satisfactory results, repeat the search with the same parameters to ensure consistency.
22	garden	Click on relevant product options to verify details before making a purchase decision.
23	garden	Prioritize products that meet all specified criteria to avoid unnecessary browsing.
24	garden	Use clear and concise language in instructions to enhance search accuracy and efficiency.
25	grocery	Use specific dietary and product attributes in your search queries to narrow down results effectively.
26	grocery	Always include price constraints in your search to filter out options that exceed your budget.
27	grocery	If initial search results do not yield desired products, refine your search terms and try again.
28	grocery	Utilize product attributes like flavor and size to ensure the items meet your specific requirements.
29	grocery	Click through search results methodically, checking for the correct product attributes before making a purchase.
30	grocery	If you encounter irrelevant results, return to the search and adjust your keywords for better accuracy.

Table 11: The WebShop memory (`data/pools/webshop1010_insights.jsonl`): 30 rules, 524 tokens, which GPT-4o-mini distilled from its own successful training sessions. All 30 are inserted.